



## Deploying

Given a scenario, describe approaches to handle pre and post-deployment steps, including items not supported via the APIs.

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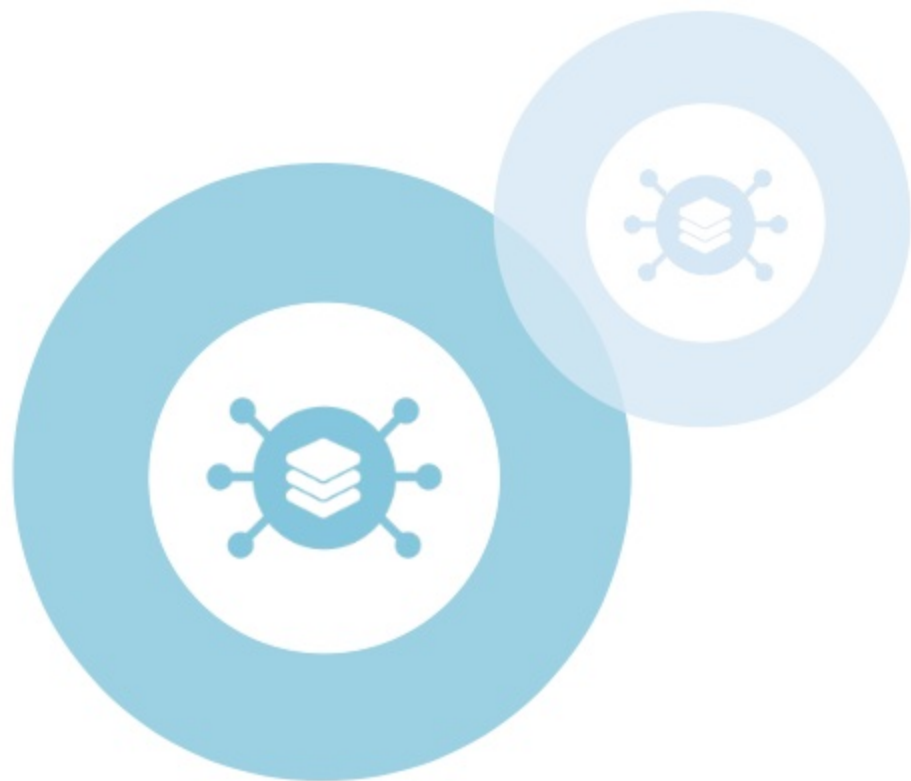
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## After studying this topic, you should be able to:

-  Identify various pre-deployment steps that should be performed prior to a deployment.
-  Identify various post-deployment steps should be performed after a deployment.
-  Describe approaches to handle pre and post-deployment steps for the given business requirements.



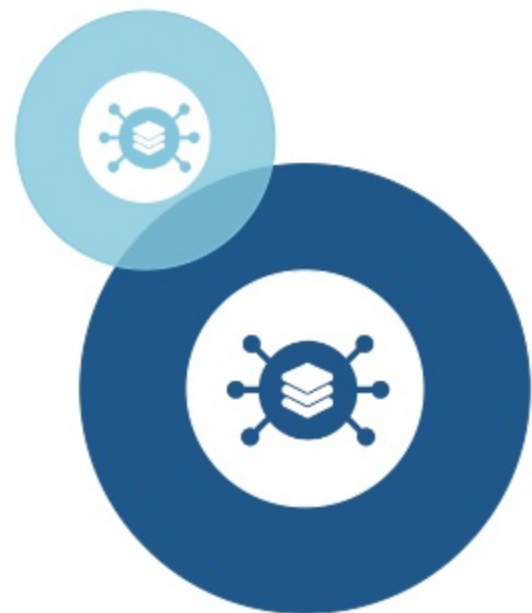
# Introduction

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Salesforce deployments often require pre and post-deployment steps. **Pre-deployment steps** include important activities such as planning, sandbox testing, documentation of manual steps, freezing users, etc. It is necessary to perform these steps to ensure that no issues arise during or after the deployment.

**Deployment activities** include choosing a suitable tool, performing deployment validation, identifying the correct order of dependencies, and performing the deployment.

**Post-deployment steps** are necessary to make sure that the deployed components work as expected. Some important steps include reactivating deactivated features, such as validation rules and email deliverability, performing data migration in the correct order, monitoring the deployed features, etc.

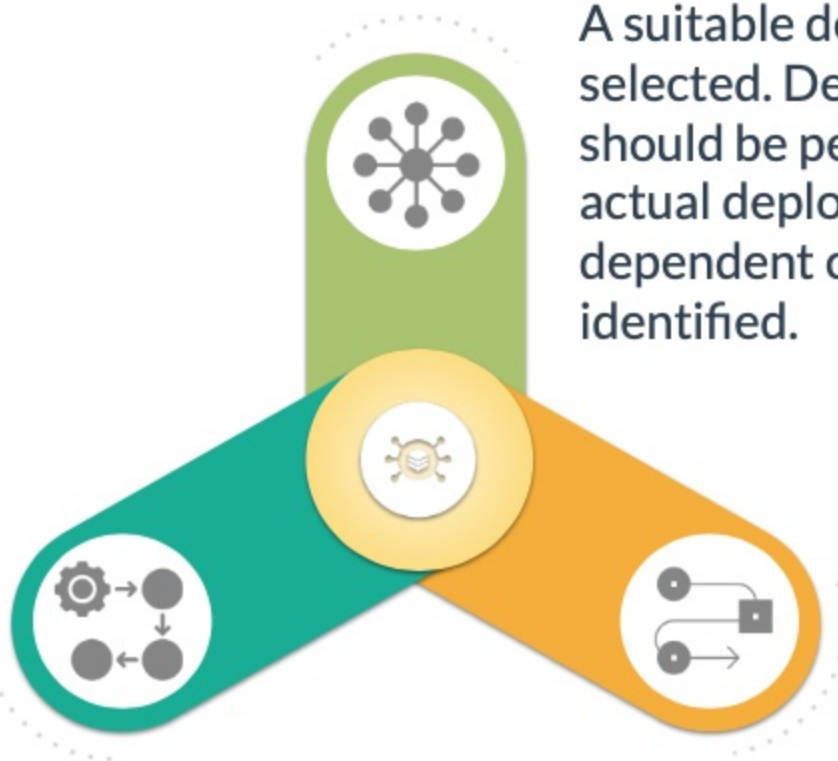




# Deployment Steps

## Pre-Deployment Steps

Pre-deployment steps include planning, checking the audit trail, backing up data and metadata, documenting manual steps, disabling automation, etc.



## Deployment Steps

A suitable deployment tool should be selected. Deployment validation should be performed prior to the actual deployment. The order of dependent components should be identified.

## Post-Deployment Steps

Post-deployment steps include post-deployment testing, reactivation of features that were deactivated, unfreezing users, re-enabling email deliverability, etc.

# Deployment Steps

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# Pre-Deployment Steps

Pre-deployment steps are **activities** that are performed **before** a Salesforce deployment.



## ❖ PLANNING

A Salesforce deployment should be **well-planned**. The deployment implications should be considered. Various **positive and negative scenarios** should be built and tested. All users should be **notified** and **frozen** or **asked not to log in** during the deployment timeframe.

## ❖ TESTING

A sandbox that is a **true copy/replica** of the production environment should be used to test the application before deployment.

## ❖ AUDIT TRAIL

The audit trail in the production environment should be **checked** to ensure that **no changes** were made that would be overwritten.

# Pre-Deployment Steps

Pre-deployment steps are **activities** that are performed **before** a Salesforce deployment.



## ❖ DEPENDENCIES

Dependencies should be **checked** and all **related components** should be included in the package.

## ❖ BACKUP

**Data and metadata** should be **exported/backed up**, especially for components that are being replaced or modified.

## ❖ MANUAL STEPS DOCUMENTATION

Any manual steps required, such as **data migration** and **feature activations**, should be **documented** prior to the deployment.



# Pre-Deployment Steps

Pre-deployment steps are **activities** that are performed **before** a Salesforce deployment.



## ❖ AUTOMATION

**Validation Rules, Flows, and Apex Triggers** that would prevent the deployment of new records or changes should be **deactivated**. Email deliverability should be **disabled** to ensure that mass changes **do not trigger system emails and notifications**.

## ❖ TEST SCRIPTS & TRAINING

**Test Scripts** to run after the deployment should be prepared. **Training sessions** should be organized for users in a sandbox environment.

## ❖ UNSUPPORTED COMPONENTS

Components **not supported by the API**, such as sharing settings and organization-wide email address, should be **identified**, and their **manual deployment** should be planned.

# Deployment

Deployment steps are **activities** that are performed **during** a Salesforce deployment.

## ❖ DEPLOYMENT TOOL

A suitable deployment tool, such as **Change Sets** or **Metadata API**, should be selected. The **deployment package** should be prepared accordingly.

## ❖ DEPLOYMENT VALIDATION

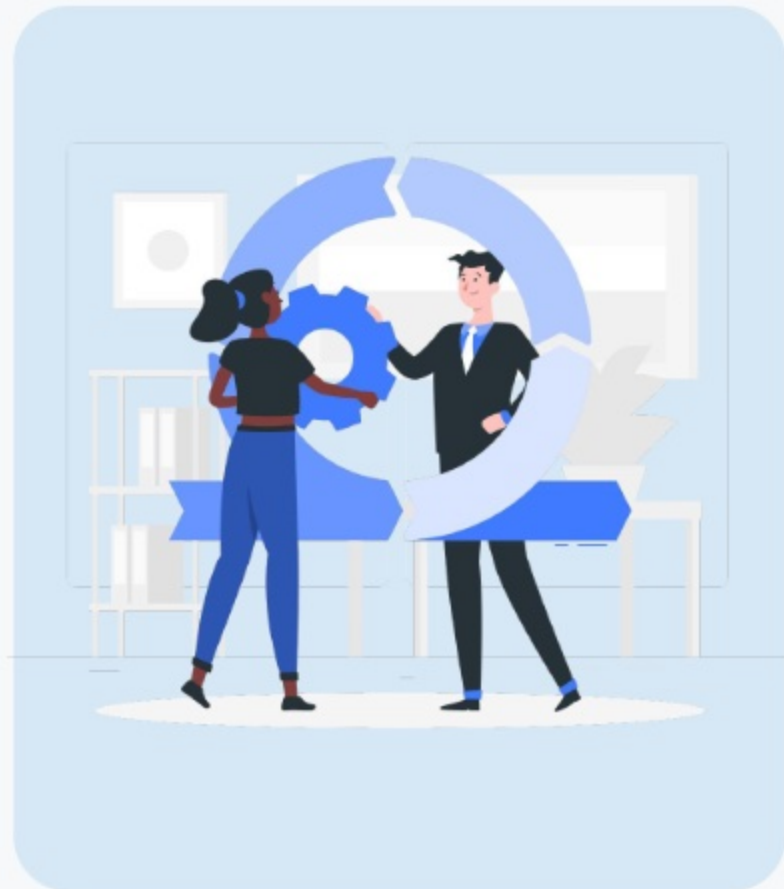
Before performing the actual deployment, it should be **validated**. For example, when **deploying a change set**, it should **first be validated** to view the **success or failure messages** during the actual deployment.

## ❖ DEPLOYMENT ORDER

The correct order of deployment should be identified for the components with **dependencies** to ensure their successful deployment.

## ❖ DEPLOYMENT EXECUTION

The deployment should be performed using the **prepared deployment package** and the **selected deployment tool**.





# Post-Deployment Steps

Post-deployment steps are **activities** that are performed **after** a Salesforce deployment.

## ❖ VALIDATION

**Post-deployment testing** should be performed to ensure that all **deployed components** and **their interaction with existing components** work as expected in the production environment.

## ❖ REACTIVATION

Any features, such as validation rules, flows, and triggers, that were **deactivated** before the deployment should be **reactivated**. User accounts should be **unfrozen**. **Email deliverability** should be **enabled** to ensure that users can receive emails again.

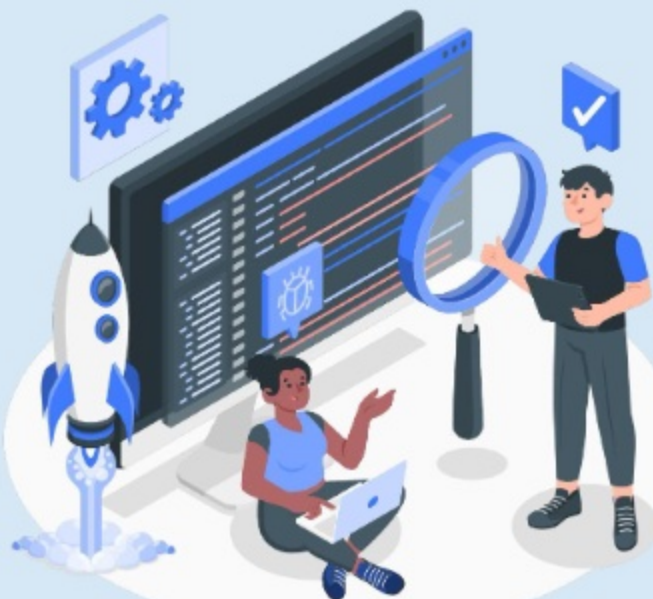
## ❖ MANUAL CHANGES

Any configuration changes that could **not be automated** via the API should be **performed manually**.



# Post-Deployment Steps

Post-deployment steps are **activities** that are performed **after** a Salesforce deployment.



## ❖ DATA MIGRATION

**Data migration or updates** should be performed if the deployment involves **new fields or changes to data structures**.

## ❖ MONITORING & SUPPORT

The deployed features should be **monitored** to ensure that there are **no issues**. **Support** should be provided to **end users who encounter issues**.

## ❖ ADDITIONAL STEPS

**System documentation** should be updated to ensure that it reflects the deployed changes. Changes should be **communicated** to end users and stakeholders. **Training** should be provided if necessary.



# Business Scenario

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## Business Scenario

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Cosmic Green Innovations, a global leader in sustainable energy solutions, leverages Salesforce to manage its extensive client relations, project management, and analytics. The company's offerings range from renewable energy projects to environmental impact assessments, requiring sophisticated Salesforce customizations to meet diverse client needs.

With operations spanning North America and Europe, the company utilizes two primary Salesforce development teams. Despite their expertise, performing complex deployments across teams and ensuring consistent performance has been challenging. To streamline deployment processes and enhance application reliability, the company's Salesforce architect needs to analyze the existing deployment processes and recommend pre and post-deployment steps to improve the overall efficiency and consistency of Salesforce deployments.

# Requirements & Solutions

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# Requirement 1



## SCENARIO

The development team in North America has been working on a new custom application for managing clients, sales opportunities, and energy solutions. The application utilizes accounts, contacts, leads, opportunities, opportunity line items, and products, which need to be migrated from an external system. In addition, it uses certain metadata components, such as an organization-wide email address, that are not supported by the Metadata API. The architect needs to recommend appropriate approaches to handle the deployment process, including the pre and post-deployment steps and the data migration order.



# Solution 1



## SOLUTION

Prior to the deployment, it is a best practice to **document the manual deployment steps** for any **components that cannot be deployed automatically**, such as the steps required for setting up the organization-wide email address required by the application. A suitable deployment tool, such as **Metadata API**, should be utilized for deploying the application components. Once the deployment is complete, the manual deployment steps should be performed. This should be followed by the **data migration of records in this order** to ensure that parent records are loaded before the related records: **Accounts, Contacts, Leads, Products, Opportunities, and Opportunity Line Items**



## Requirement 2



### SCENARIO

The European development team is working on a custom data analytics application that focuses on customer engagement data. The application contains several custom dashboards with multiple running users. The team is planning to use Change Sets for the deployment of the application and would like to avoid possible delays during deployment. After the deployment, thousands of customer accounts, contacts, and cases will be migrated from an on-premises system. The architect must provide deployment-related considerations and identify the key areas that must be tested as part of the data migration process.



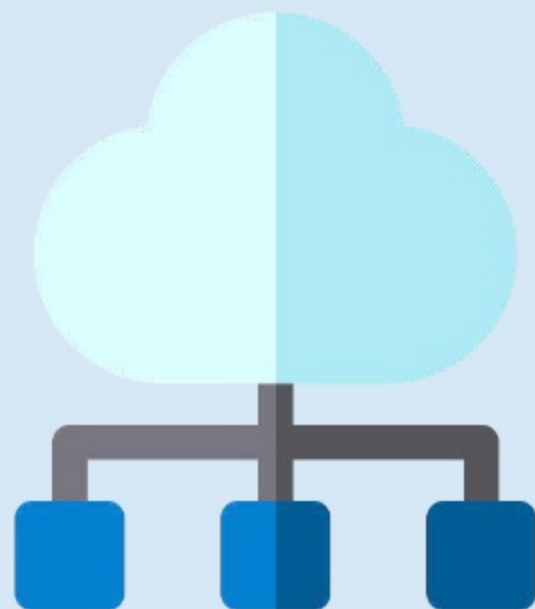


## Solution 2 (Part 1)



### SOLUTION

A few days before the planned deployment date, the **Change Sets** should be **validated** to view any success or failure messages that would occur during the actual deployment. After the validation, a **quick deployment** can be performed. For the deployment of the custom dashboards, it is important to note that **user fields** are **preserved** during a metadata deployment. Salesforce attempts to **locate a matching user** in the target org by comparing usernames during the deployment. If a **username in the source org** does not exist in the **target org**, **Salesforce displays an error** and the **deployment stops** until the username is resolved or removed.



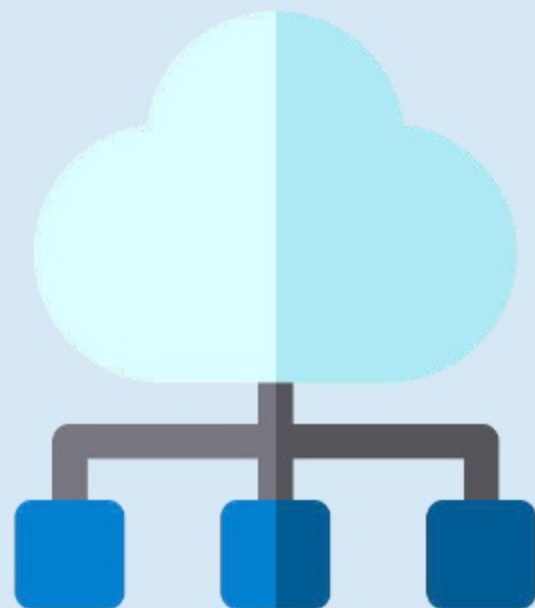


## Solution 2 (Part 2)



### SOLUTION

Some key areas that can be tested as part of the data migration include **case association** with the correct account and contact, **case ownership** along with **associated entitlements and milestones**, and **data transformation** against the on-premises system.



## Learn More

-  [Understanding Metadata API](#)
-  [Change Set Development Model](#)
-  [Maintaining User Preferences](#)
-  [Deploy Your Changes](#)
-  [Salesforce Deployment Checklist](#)

